POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2019
Programme: BE | Full Marks: 100
Course: Chemistry | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define buffer solution. Explain the mechanism of acidic buffer solution. Calculate the pH of 500 ml of buffer solution containing 0.2M of ammonium sulphate and 0.3M ammonium hydroxide which is 2.1% ionized in dilute solution. 7
(K_b of ammonium hydroxide is 1.8×10^{-5})

   b) Define electrochemical cell. The standard reduction potentials of Cu^{++}/Cu and Ag^{+}/Ag electrodes are 0.34V and 0.80V respectively. Construct a galvanic cell using these E^0 values. For what concentration of Ag^+ ions will the emf of the cell at 25^0C be zero if the concentration of Cu^{++} is 0.01M. 8

2. a) Define Ionization energy. What are the factors which affect it? Discuss the variation of I.E in the periodic table. 7

   b) Give reason: 8
      i) $TiCl_3$ is colored but $TiCl_4$ is colorless.
      ii) Zn^{+2} salts are white.
      iii) Cu is not considered as transition element.
      iv) Transition elements show variable oxidation state.

3. a) Define carbocation. How are carbocation stabilized? Write any two methods of their formation. 7

   b) What is β-elimination reaction? Write the differences between SN_1 and SN_2 reaction. 8

4. a) Write notes on 7
      i. Natural Rubber
      ii. Nylon-6,6

   b) Define the term polymerization. Explain the free radical mechanism for the polymerization of ethene. 8

5. a) How is cement manufactured? Explain in brief. 7

   b) Write principle and important applications of mass spectrometry in analytical field. Write the application of TLC. 8

Testpage

6. a) What is water pollution? Point out the major causes of water pollution, their adverse effects and possible remedies. 7

   b) Define alkalinity. How is it estimated in the laboratory, explain? 8

7. Write short notes on: (**Any two**) 2×5
   a) Aromatic electrophilic substitution reaction
   b) Chromatography
   c) Silicon polymer